



University of Fort Hare
Together in Excellence

PROCUREMENT DOCUMENT

FOR

HVAC INSTALLATION

FOR

FORT HARE UNIVERSITY FINANCE OFFICE

TENDER NUMBER: _____

NOVEMBER 2025

(RETURNABLE DOCUMENT – SINGLE VOLUME)

<u>Issued By:</u>	<u>Prepared By:</u> CA du Toit Eastern Cape (Pty) Ltd 22 Shirley Street, Newton Park Gqeberha, 6001
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NAME OF TENDERER: _____

The Offered Total/s of the Prices inclusive of value added tax is:

Total from FINAL SUMMARY _____

Tenderer's CIDB CRS number _____

NOTE: The above is for recording of tender information purposes only. The Tenderer is required to fully complete C1.1: FORM OF OFFER AND ACCEPTANCE, which formally constitutes the tender offer submitted by the Tenderer, and failure to do so, could result in disqualification of the tender

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PART T1.5 – SCHEDULE OF EQUIPMENT OFFERED

Technical particulars and trademarks of all equipment offered in the tender (including equipment that has not specifically been specified, but will form part of the installation), must be supplied in this schedule.

The following information shall be provided in full and returned with the contractors' tender. Failure to comply with this condition of tender may invalidate the tender.

Tenderers shall attach to this page technical datasheets and product literature of the equipment offered (details to be provided for each individual equipment ID (i.e. AC 1, AC 2, EX 1, FA 1, etc):

AHU: Make, Model, Physical Size, Coil Selection and Fan Selection

VRV Condenser: Make, Model, Physical Size, Capacity.

Split Units: Make, Model, Capacity, Refrigerant & Compressor type (ie.inverter)

Fans: Make, Model, Duty point on Selection Curve, Type of Fan, Noise Level

Diffusers and Grilles: Make, Model & Material Specification

PART C2: PRICING DATA

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PART C2.1 – PRICING INSTRUCTIONS

2.1.1 PREAMBLE TO THE SCHEDULE OF PRICES

2.1.1.1 A lump sum price shall be provided.

2.1.1.2 The pricing schedule shall be used to evaluated progress claims.

2.1.1.3 The Contractor shall be required to provide a detailed cost breakdown of the individual lump sum prices to enable re-measurement of any changes and or/ variations.

2.1.1.4 All prices or rates inserted in the Pricing Schedule shall be EXCLUDING VAT. Provision has been made on the Summary Page, of the Pricing Schedule, for the addition of VAT.

2.1.1.5 Arithmetical errors of responsive tenders will be corrected in the following manner:

- Where there is a discrepancy between an amount shown in figures, and the corresponding amount stated in words, the amount stated in words shall take preference.
- In the Pricing Schedule, if there is an error in the line item total resulting from the product of the quantity and the unit rate, the line item total shall govern, and the rate shall be corrected.
- Where there is a misplacement of the decimal point in the unit rate, the line item total shall govern and the unit rate will be corrected.
- Where there is an error in the total of the prices either as a result of other corrections required by this checking process or in the tenderer's addition of prices, the total of the prices shall govern and the tenderer will be asked to revise selected item prices (and their rates in the Pricing Schedule) to achieve the tendered total of the prices.

Should a tenderer be unwilling to make the corrections ordered by the Engineer, the tender may be disqualified.

2.1.1.6 The units of measurement described in the Pricing Schedule are metric units. Abbreviations used in the Pricing Schedule are as follows :

mm	=	millimetre	hrs	=	hours
m	=	metre	kg	=	kilogram
km	=	kilometre	t	=	ton (1 000 kg)
m ²	=	square metre	No.	=	number
m ² .pass	=	square metre-pass	sum	=	lump sum
ha	=	hectare	MN	=	MegaNewton
m ³	=	cubic metre	MN.m	=	MegaNewton-metre
m ³ .km	=	cubic metre-kilometre	P C sum	=	Prime Cost sum
ℓ	=	litre	Prov sum	=	Provisional sum
kℓ	=	kilolitre	%	=	per cent
MPa	=	MegaPascal	kW	=	kilowatt

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PART C2.2 – SCHEDULE OF PRICES

SCHEDULE 1: PRELIMINARY AND GENERAL ITEMS

Item	Description	Unit	Qty	Rate	Amount
1	Preliminary and General Items				
1.1	Compliance with the conditions of contract, contract data, preliminaries and general and all administrative costs and overheads.	Sum	1		R
1.2	Insurances	Sum	1		R
1.3	Construction Guarantee	Sum	1		R
1.5	Liaison with Main Contractor on programming and execution of works.	Sum	1		R
1.6	Liaison with other Contractors and Service providers (ie., Electrician, Medical Gas, Plumber, etc.) for final co-ordination between services with site management to ensure timeous and correct installation.	Sum	1		R
1.7	Contract Management and Site Supervision	Sum	1		R
1.8	Nett price to comply with all the requirements of the Health & Safety specification, ie Health and Safety file, Medicals, PPE for the duration of the contract.	Sum	1		R
1.9	Compilation of detail Operating and Maintenance Manuals, as-built drawings and documentation as specified and to the satisfaction of the Engineer. 3-off printed copies and 1x electronic version.	Sum	1		R
1.10	Training of the Employers staff.	Sum	1		R
1.11	12 months guarantee from date of Practical Completion Certificate.	Sum	1		R
1.13	12 months full maintenance and service of the installation in accordance with manufacturer's and suppliers specifications.	Sum	1		R
1.14	Any other P&G items: Contractor to specify	Sum	1		R
	TOTAL SCHEDULE 1 : CARRIED FORWARD TO FINAL SUMMARY				R

SCHEDULE 2: HVAC EQUIPMENT INSTALLATION

Item	Description	Unit	Qty	Rate	Amount
2.1	<u>VENTILATION SYSTEMS</u> <i>Supply and Install as per Tender Drawings P8080M02-001</i>				
2.1.1	FA 1 - complete as specified and indicated on the drawings				
2.1.1.1	Fan Assembly - 260 L/s @ 200 Pa ESP	no	1		R -
2.1.1.2	2D sound attenuators	no	2		R -
2.1.1.3	System ductwork and air diffusion - complete	sum	1		R -
2.1.1.4	Installation of complete system	no	1		R -
2.1.1.5	Testing and Commissioning.	sum	1		R -
2.1.2	FA 2 - complete as specified and indicated on the drawings				
2.1.2.1	Fan Assembly - 280 L/s @ 200 Pa ESP	no	1		R -
2.1.2.2	2D sound attenuators	no	2		R -
2.1.2.3	System ductwork and air diffusion - complete	sum	1		R -
2.1.2.4	Installation of complete system	no	1		R -
2.1.2.5	Testing and Commissioning.	sum	1		R -
2.2	<u>VRF SYSTEMS</u> <i>Supply and Install as per Tender Drawings P8080M02-001</i>				
2.2.1	VRF 1 - complete as specified and indicated on the drawings Supply and Delivery of:				
2.2.1.1	VRF outdoor unit, heat recovery type, 56kW min cooling capacity	no	1		R -
2.2.1.2	Ceiling Cassette units, 600x600, 2.8kW min cooling capacity, c/w wired controller	no	3		R -
2.2.1.3	Ceiling Cassette units, 600x600, 3.6kW min cooling capacity, c/w wired controller	no	1		R -
2.2.1.4	Ceiling Cassette units, 600x600, 5.6kW min cooling capacity, c/w wired controller	no	1		R -
	Carried forward				R -

Item	Description	Unit	Qty	Rate	Amount
	Brought forward				R -
2.2.1.5	Ceiling Cassette units, 600x600, 7.1kW min cooling capacity, c/w wired controller	no	5		R -
2.2.1.6	MCU units - 4 port	no	1		R -
2.2.1.7	MCU units - 6 port	no	1		R -
	<u>Refrigeration piping and associated ancillaries</u>				
2.2.1.8	1 1/4" nominal Size	m	32		R -
2.2.1.9	7/8" nominal Size	m	32		R -
2.2.1.10	3/4" nominal Size	m	32		R -
2.2.1.11	5/8" nominal Size	m	78		R -
2.2.1.12	1/2" nominal Size	m	42		R -
2.2.1.13	3/8" nominal Size	m	90		R -
2.2.1.14	1/4" nominal Size	m	42		R -
2.2.1.15	Y-Joint / Refnet	no	2		R -
2.2.1.16	Condensate drains, complete with insulation	sum	1		R -
	Installation of:				
2.2.1.17	VRF outdoor unit, heat recovery type, 56kW min cooling capacity	sum	1		R -
2.2.1.18	Ceiling Cassette units, 600x600, 2.8kW min cooling capacity, c/w wired controller	sum	1		R -
2.2.1.19	Ceiling Cassette units, 600x600, 3.6kW min cooling capacity, c/w wired controller	sum	1		R -
2.2.1.20	Ceiling Cassette units, 600x600, 7.1kW min cooling capacity, c/w wired controller	sum	1		R -
2.2.1.21	MCU units - 4 port	sum	1		R -
2.2.1.22	MCU units - 6 port	sum	1		R -
2.2.1.23	Refrigeration piping and associated ancillaries	sum	1		R -
2.2.1.24	uPVC insulated condensate drains	sum	1		R -
2.2.1.25	Additional refrigerant charge	kg	15		R -
2.2.1.26	Testing and Commissioning	no	1		R -
	Carried forward				R -

Item	Description	Unit	Qty	Rate	Amount
	Brought forward				R -
2.3	<u>CABLE LADDERS AND WIRE WAYS</u>				
	Refrigerant piping to be contained in cable ladder or wire basket. Descriptions for the trays or basket required for the mechanical installation are deemed to include fixing in position and are also deemed to include waste pieces which cannot be re-used.				
2.3.1	150mm Medium duty welded wire mesh	m	150		R -
2.3.2	Elbow	no	20		R -
2.3.3	Tee	no	2		R -
2.3.4	Riser/dropper	no	4		R -
2.4	<u>CORROSION TREATMENT</u>				
2.4.1	All Condensing units to be corrosion treated with Bluchem				
2.4.1.1	VRF Condensing units	no	1		R -
2.4.1.2	On site touch up	sum	1		R -
2.5	<u>ELECTRICAL</u>				
	<i>Electrically connect, test and verify power supply to the following equipment as specified</i>				
2.5.1	Indoor Units	no	10		R -
2.5.2	Condensing Units	no	1		R -
2.5.3	Electrical COC for the HVAC installation	sum	1		R -
2.6	<u>CONTROLS & CONTROL WIRING</u>				
	<i>Supply and install, including connection, control wiring between Control Panels, field devices and equipment. Wiring to run in 20mm PVC conduit with flexible connections and termination glands to each device/terminal.</i>				
2.6.1	Wired remote controllers	no	10		R -
	Carried forward				R -

Item	Description	Unit	Qty	Rate	Amount
	Brought forward				R -
2.7	<u>DUCTING</u>				
	<i>Supply and Installation of galvanised sheet metal ducting</i>				
2.7.1	Diameter 355 mm spiral ducts	m	4		R -
2.7.2	Diameter 275 mm spiral ducts	m	6		R -
2.7.3	Diameter 250 mm spiral ducts	m	5		R -
2.7.4	Diameter 200 mm spiral ducts	m	18		R -
2.7.5	Diameter 150 mm spiral ducts	m	30		R -
2.7.6	Diameter 100 mm spiral ducts	m	7		R -
2.8	<u>DUCT FITTINGS</u>				
	<i>Supply and Installation of galvanised sheet metal duct fittings</i>				
	<u>Spigot</u>				
2.8.1	Diameter 200 mm	no	1		R -
2.8.2	Diameter 150 mm	no	4		R -
2.8.3	Diameter 100 mm	no	4		R -
	<u>Bend</u>				
2.8.4	Diameter 355 mm	no	1		R -
2.8.5	Diameter 200 mm	no	1		R -
2.8.6	Diameter 150 mm	no	1		R -
	<u>Taper Piece / Reducer</u>				
2.8.7	Diameter 355 mm - 275 mm	no	1		R -
2.8.8	Diameter 355 mm - 250 mm	no	1		R -
2.8.9	Diameter 275 mm - 200 mm	no	1		R -
2.8.10	Diameter 250 mm - 200 mm	no	1		R -
2.8.11	Diameter 200 mm - 150 mm	no	3		R -
2.8.12	Diameter 150 mm - 100 mm	no	6		R -
	Carried forward				R -

Item	Description	Unit	Qty	Rate	Amount
	Brought forward				R -
2.9	SUNDRY ITEMS				
2.9.1	Hanger bracket to securly support and fix all the equipment, ducting, wire baskets and refrigerant piping	sum	1		R -
2.10	Any other items: Contractor to specifiy				
2.10.1		sum			
2.10.2		sum			
2.10.3		sum			
	TOTAL FOR SCHEDULE NO. 2 :CARRIED FORWARD TO FINAL SUMMARY				R -

FINAL SUMMARY - MECHANICAL HVAC INSTALLATION

Schedule:

1	Subtotal Bill No: 1 - Preliminary and General	R	-
2	Subtotal Bill No: 2 - HVAC Equipment and Materials	R	-

Sub-Total:	R	-
Add : 5% Contingency	R	-
Sub-Total	R	-
15 % VAT	R	-
TOTAL TENDER PRICE	R	-

NOTES TO BE READ IN CONJUNCTION WITH THE TENDER PRICE.

- 1 The amount given as the "Total Tender Price" and carried forward to Part C1.1 Form of Offer and Acceptance will be regarded as including the Provisional and Contingency amounts shown in the various schedules and no adjustments will be made for any failure by Tenderer's to include for any pricing items, sums or amounts in the total appearing on the official offer.
- 2 The entire mechanical installation in this contract shall be completed prior to completion and acceptance.
- 3 All Material offered should, wherever possible, be of South African manufacture with strong local (Port Elizabeth) after-sales service and backup. In the final adjudication of Tenders, preference will be given to bids offering such equipment and materials.

Name of Mechanical Contractor

Signature of Authorised Representative of Mechanical Contractor

Date

PART C3: SCOPE OF WORK

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PART C3.1 – GENERAL TECHNICAL SPECIFICATION

The Standard Technical Specification for Air Conditioning and Ventilation Installations of the National Department of Public Works of 1998, Issue XII, shall apply. This document is not included, but can be obtained from the Department's website – www.publicworks.gov.za.

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PART C3.2 – PROJECT TECHNICAL SPECIFICATION

1. SCOPE OF WORK

The scope of work is summarised below and includes, but not limited to the demolition and removal from site redundant HVAC plant and the supply, delivery, installation, testing and commissioning of new Air Conditioning Installations for the Sanlam Lecture Venue.

Demolition and Removal of Existing Installation

- The existing chilled water installation shall be de-commissioned, dismantled and removed from site.

New HVAC Installations

- Five (5) ducted split systems
- Filter plenum boxes to suite
- New and modified GMS ducting as indicated on the drawings
- New electrical supplies to the ducted split systems.
- Upgrading of the Control system and BMS

Associated Builders' Work

- All builders' work required shall be included and allowed for.

Associated Electrical Work

- The contractor shall supply and install one (1) new control panel, **CP1**. The existing electrical supply shall be re-used.
- Weatherproof isolators shall be provided within 3m of all AC outdoor units.
- The Contractor shall allow for the electrical connection of each AC unit.
- The Contractor shall allow for the electrical cabling and, terminations to from Isolator to individual condensing units. In addition to this, the Contractor shall also allow for all electrical and control wiring between the condensing units and indoor units, motion sensors and controls wiring and instrumentation.
- Individual systems will be switched On/Off via remote (and illuminated) push button with adjustable rundown timer from inside each lecture room.
- CO₂ control shall be provided via duct mounted return air CO₂ sensors controlling modulating fresh air intake dampers on each system.
- The Contractor shall allow for wire basket containment of all electrical and control wiring and connection of all associated mechanical equipment to ensure fully functional and compliant systems.
- The BMS shall be upgraded to incorporate the installation changes.

2

GENERAL

Where reference is made to Tenderer or Contractor in this specification the intent is to identify the specialist Heating, Ventilation and Air Conditioning Mechanical Contractor.

2.1

Statutory and Regulatory Requirements

The Installation shall be designed, erected, commissioned and maintained in compliance with the following applicable Acts and Regulations, as applicable:

- (a) SANS 0142 - 1978: Code of Practice for the Wiring of Premises, as amended.
- (b) The Occupational Health and Safety Act of 1993, as amended.
- (c) Mines and Works Regulations, Government Notice Number R1609 of the 28th September, 1962 as amended
- (d) SANS 10400 - 2011: The application of the National Building Regulations.
- (e) Local Municipal by-laws and regulations as well as the regulations of the Local Electrical Supply Authority and Health Department
- (f) Local Fire Regulations
- (g) Aviation Authority
- (h) Construction Regulations and Health and Safety Requirements

In addition the contractor shall exempt the Employer from all losses, costs or expenditures which may arise as a result of the contractor's negligence to comply with the requirements of the regulations enumerated in this Clause.

It shall be assumed that the contractor is conversant with the above-mentioned requirements. Should any requirement, by-law or regulation, which contradicts the requirements of this Document, apply or become applicable during erection of the Installation, such requirements, by-law or regulation shall overrule this Document and the contractor shall immediately inform the Engineer of such a contradiction.

All equipment and installations shall comply with the requirements of the Occupational Health and Safety Act 85 of 1993 and relevant SANS specifications. Where conditions are at variance, the Project/Detail Technical Specification shall have preference over both General Technical Specifications and the Drawings.

Under no circumstances shall the Contractor carry out any variations to the installation in terms of such contradictions without obtaining the written permission to do so from the Engineer.

The contractor shall supply and install all notices and warning signs that are required by the appropriate laws or regulations and/or by these Documents.

2.2

Reference Specifications and Standards

The latest revision of any General Technical Specifications referred to in this Specification will be applicable.

Where a specification or standard is not specifically referred to it will be assumed that the relevant SANS, ISO, BSS, DIN or equivalent American standard, listed in order of preference, will apply.

The SI ("Le Système International d' Unités") - Metric System of Units will apply. Refer to SANS - M33A: The International Metric System: Guide to the use of the SI in South Africa.

All equipment and installations shall comply with the requirements of the Occupational Health and Safety Act 85 of 1993 and relevant SANS specifications.

Where conditions are at variance, the Detail Technical Specification shall have preference over both General Technical Specifications and the drawings.

2.3 General Design Conditions

Environmental Conditions

Summer	:	28,0 °C db.
	:	22,0 °C wb.
Winter	:	4,3 °C db.
	:	4,1 °C wb.

Room Condition

Summer	:	23°C ± 1°C
	:	50%rh ± 10%rh
Winter	:	21°C ± 1°C
	:	50%rh ± 10%rh

The temperature setpoint shall be adjustable between 19°C and 25°C, unless otherwise required or specified.

Ventilation Rate

Fresh Air (minimum)	:	10,0 l/s per person
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Where CO₂ sensing is utilised the maximum permissible concentration levels shall be maintained between 700ppm and 900ppm.

Noise Levels

The following noise levels shall not be exceeded:

35 NC, 1,8m above floor level in the centre of the occupied offices and general space.

NC 30, 1,8m above floor level in the centre of any Medical/Procedure/Ward space.

Internal Loads

The internal heat load of lighting shall be taken as 10 W/m².

Altitude

The site is situated just above sea level (±50m) and all capacities indicated in this specification and on the drawings are relevant to this altitude.

2.4 Guarantee Performance

The contractor shall guarantee that the installation will be installed and set in such a way that, depending on the capacities as specified, the environment in the air conditioned areas are maintained according to the design conditions as specified.

The contractor shall also guarantee:

- That the installation shall function without excessive draughts, noise or vibrations.
- That equipment shall be installed, commissioned and set properly in order to attain the design conditions specified.
- That an installation will be erected which will function in a safe and satisfactory manner in every respect.

Any decisions or instructions in this regard by the Engineer shall be final.

2.5 Material and Workmanship

All materials shall be of the qualities specified and the contractor shall, upon request of the Engineer, furnish him with proof to his satisfaction that the materials are of the specified quality.

The contract works shall be executed in a workmanlike manner, to the satisfaction of the Engineer, and shall include all materials and equipment required for the successful operation of the installation as specified.

All materials and equipment used for the installation shall be new and undamaged.

The contractor shall, if requested by the Engineer, provide samples of material and equipment for approval. Such samples, if judged necessary by the Engineer, may only be returned after completion of the installation in order to ensure that the quality of the installed product is the same as that of the approved sample.

The contractor shall use only new materials of the best quality. A high standard of workmanship shall be maintained. The Engineer shall have the right to approve or disapprove any part of the installation if he deems it necessary in accordance with his professional judgement, interpretation of the Specification and accepted practice.

The contractor will be notified of any such disapproval of a part or parts of the installation and he will immediately rectify or replace that specific part or parts of the installation to the satisfaction of the Engineer. Any such instruction will not be deemed contrary to the Contract and will not relieve the contractor of his responsibility to complete the installation on the originally agreed completion date.

2.6 Alternative Equipment

The words "similar or equal approved" is always implied whether following the description of equipment, or not.

Should names of equipment manufacturers appear in the schedules, those are around which the design has been based and all equipment so scheduled is complementary.

The tenderer may offer alternative makes providing that such alternatives are of equal quality to those "selected" by the Engineer and should the tender be accepted the approval of the Engineer is obtained before ordering.

The Engineer shall be the sole judge of the equipment offered and his decision shall be final.

As stated above all equipment selections are complementary and it is the responsibility of the successful tenderer to ensure that: -

- His alternative offer can be housed within the space provided, leaving sufficient clearance surrounding the unit as laid down by local regulations and for ease of maintenance. Should the alternative have to be replaced by that as listed in order to meet these requirements, it shall be at no additional cost to the Contract.

- All openings in the structure meet with the requirements of his alternative offers, or ensure that timely details of any relevant amendments are requested to prevent any abortive work on site.
- If there are any monetary benefits which can be offered to the Client through an alternative item of equipment, the Tenderer must incorporate all secondary cost variations into that saving reflected (i.e. the price reduction offered for an alternative chiller shall incorporate any additional costs in upgrading a pump and/or its motor size which could result from an increase in resistance through its heat exchangers).

Further, all relevant technical data, especially regarding energy consumption, rated output, capacity, variations in resistances, etc., must also be submitted in order that the Engineer may evaluate savings in capital costs against any increases in operating costs prior to accepting that particular alternative offer.

2.7 Make of Equipment

The contractor will be allowed to offer equivalent equipment for consideration, under the following explicit conditions:

The equipment shall comply in all respects with the requirements specified and shall be subject to approval by the Engineer.

Equipment shall be of equivalent quality and performance.

The delivery period for such equipment shall not jeopardise the construction programme.

The local manufacturer or supplier shall have adequate service facilities and/or keep sufficient stocks of spares required for servicing.

The equipment shall be offered as an ALTERNATIVE if offered at tender stage with the accompanying cost saving.

The contractor shall supply such information regarding the equipment as required by the Engineer.

2.8 Noise and Vibration Control

The Installation shall operate without causing undue noise and vibration. The Contractor shall take the necessary precautions to ensure that noise levels in occupied areas as specified elsewhere in the Specification are not exceeded.

Noise generating equipment such as fans, compressors, pumps, motors etc. shall be selected to operate as close to the point of maximum efficiency as possible. It is the responsibility of the Contractor to check operating noise levels of equipment before tendering. Tenderers offering equipment with low noise ratings may receive preference.

Where it is not possible to meet the specified sound levels due to the noise generated by the equipment offered, or due to inadequacies in the building structure, such deficiencies shall be qualified in the tender.

2.9 Inspections and Testing

Inspections (Part III, SAACE - 1978)

The Engineer shall have general supervision and give direction to the contract works. Supervision shall comprise such periodic visits, as the Engineer may consider necessary to inspect the contract works for conformity with the contract documentation and to provide clarification and further information as necessary.

The Engineer shall have the power at any time to inspect and examine any part of the contract works or any materials intended for use in or on the contract works, either on the site or at any factory, workshop or other place where such parts or materials are being constructed manufactured or at any place where same are lying or from where they are being obtained, and the Contractor shall give all such facilities as the Engineer may reasonably require to be given for such inspection and examination.

The Contractor shall not be liable for the cost of inspecting materials at the place of manufacture, construction or storage nor any travelling or accommodation costs arising out of the execution of such inspection etc.

Testing

The Contractor shall supply all test equipment, test facilities and everything necessary at his cost, to perform these tests. The minimum testing and commissioning equipment required is as follows:

1. Velocity meter (hot wire anemometer).
2. Crane type manometer for balancing valves.
3. Sling psychrometer
4. Revolution counter suitable for measuring fan and motor shaft rotation.
5. Megger equipment.
6. Clamp on ammeter.
7. Voltmeter.
8. Power factor meter.
9. Ohmmeter suitable for continuity testing.
10. Neon type ON/OFF test lamp.
11. Maximum indicating ammeter suitable for measuring peak motor starting currents.

The Contractor shall record all measurements taken during testing and shall do the necessary adjustments until the Engineer is satisfied with the results.

The Engineer shall be notified one (1) week in advance of any tests so that he may witness such tests.

Unless otherwise specified, the Contractor will be required to perform the following tests:

a) Electrical Switchboards and Control Panels:

A simulated functional test in the factory to ensure the correct operation of equipment, controls, interlocks and measuring circuits.

A 2,5kV pressure test in the factory.

b) Electrical:

A megger test of all circuits prior to energising the switchboard for pre-commissioning.

Phase rotation.

c) Ducting:

Pressure test medium and high pressure ducting in terms of SANS 0173 - 1980: Code of Practice for the Installation, Testing and Balancing of the Air Conditioning Ductwork.

d) Piping:

Pressure test of chilled water piping at a test pressure of 1,5 times the maximum working pressure at the lowest point in the system, but not less than 700kPa.

All instrumentation or other equipment, which could be damaged during the pressure test, shall be removed from the pipe systems. The relevant system shall be filled with water and all high points shall be vented at least 24 hours before the test.

The duration of the pressure test shall be 2 hours, after which no water leaks shall be visible and no pressure drop shall occur after corrections have been made for changes in ambient temperature during the test period.

Pressure tests shall be completed prior to insulating or cladding of piping.

If leaks are found, welded connections shall be cut out and re-welded and screwed joints shall be dismantled, cleaned and re-connected. Rectified piping shall be re-tested.

e) Refrigerant Piping:

Factory charged systems shall first be pressure tested and thereafter be vacuum tested. Field charged systems shall be pressure tested.

Refrigerant pipes and equipment shall be tested under vacuum and indications of leaks shall not be permitted at a pressure of 0,68 kPa absolute, maintained for a period of one hour with the vacuum pump uncoupled.

Refrigerant pipes and equipment shall be tested with clean dry nitrogen with a small quantity of refrigerant for a period of 24 hours at a test pressure of 1,1 times the appropriate maximum working pressure in terms of SANS 0147 - 1978: Code of Practice for Refrigeration and Air Conditioning Installations. All connections shall be inspected for leaks by means of a sensitive leak detector and soap bubble test.

The particular vacuum and dewpoint requirements of the VRV supplier shall be provided to ensure unconditional guarantee from the equipment supplier.

f) Automatic Controls

The functionality of each control device as well as each field sensor shall be verified. Verification shall involve, but not limited to, the following:

- Sensor accuracy and calibration
- Heating and Cooling simulation
- Equipment controllability

2.8 Tender Drawings and Design Procedures

The tender design drawings are listed in Part C4 and attached in order to give an overview of the project.

These drawings provide sufficient information for pricing purposes, but do not necessarily indicate the exact position or detail the construction of equipment. Tenderers must satisfy themselves that the equipment offered by them will fit in the available space and can be positioned so that access for maintenance, cleaning, repair or removal is possible.

NOTE: All final dimensions must be determined on site before commencing the manufacture of any equipment. It is the responsibility of the contractor to verify all dimensions on site.

The contractor shall produce, and submit for approval, comprehensive, detailed, dimensional manufacturing (workshop) drawings of all sub-assemblies and components.

The contractor shall supply two (2) paper prints of all workshop drawings. The contractor shall allow one (1) week for drawing approval. The Contractor shall schedule design review session(s) with the Engineer and the Employer to facilitate input and/or clarification of the proposed detailed design. The Contractor shall update the drawings which shall then be submitted for signature. This will ensure that all prints used for construction will be certified as approved.

Any work done by the contractor without approved and signed drawings shall be completely at his own risk.

The contractor shall update all drawings on an on-going basis. Three (3) sets of final paper prints plus a CD copy of all the drawings in AutoCAD format, shall be supplied to the Engineer as part of the Operating and Maintenance Manual.

2.9 As-Built Drawings and Operation Instruction and Maintenance Manuals

- a) All manuals One (1) set shall be bound properly and thoroughly in a durable hard plastic cover. The contents of the manual shall be clearly legible, well-structured and supplied with an index. One (1) set of PDF format, shall also be supplied on a memory stick.
- b) No Completion Certificate shall be issued before the hard copy of the all drawings, operator's manuals and maintenance manuals have been supplied to and approved by the Engineer.
- c) Concept issues of final drawings and manuals shall be ready and available with the first take over and commissioning and no take over shall be considered without such drawings and manuals.

It is stressed that the Completion Certificate will NOT be issued unless the O&M manuals are complete and approved by the Engineer.

2.10 Training

The Tenderer shall include for detail instruction and training sessions with the Employer's technical department.

2.11 Maintenance and Servicing during the 12 Month Guarantee Period

The contractual guarantee period on faulty equipment, materials and workmanship shall be twelve (12) months. The guarantee period on material, equipment and labour commences on the date of the completion certificate issued by the Engineer and expires twelve (12) months later.

During the guarantee period, the contractor shall be responsible for the complete maintenance and servicing of equipment and plant according to the suppliers/ manufacturer's specifications. Maintenance of the installation shall mean the regular servicing, lubrication, repairing, cleaning and adjustment of the installation as recommended by the manufactures as well as the free of charge replacement of any defective components during the guarantee period.

A suitably qualified and trained person shall routinely and regularly examine and test the installation once every month and shall also perform all the necessary maintenance tasks to ensure smooth and faultless operation. A quarterly report shall be submitted to the Engineer.

The contractor shall immediately, on the day of first call-out, attend to breakdown/emergency calls. In the event of non-performance by the contractor in this respect, the owner shall be entitled to make such other arrangements as are necessary, the cost of which shall be for the contractor's account or deductible from any outstanding retention monies.

In the event of a call-out by the Engineer during the one (1) year maintenance period the contractor shall make available a qualified technician to attend to any plant related problems within 4-hours (from time of instruction by Engineer to arrival on site). The Engineer may require emergency response in which case the Contractor shall take such action that is necessary to ensure an immediate response. The fault(s) shall be corrected and a report issued to the Engineer on the exact detail of the breakdown and the corrective steps taken to prevent re-occurrence of a similar fault.

A logbook shall be kept and all servicing and repairs shall be recorded in this logbook with meticulous care. The logbook shall at all times be put at the disposal of the Engineer. The contractor shall issue the logbook with full record of all services and repairs to the owner after the guarantee period has expired.

At each routine and quarterly service, the Contractor shall notify the Employer's technical service department to facilitate a staff member to accompany the service crew to validate scheduled service tasks are performed.

2.12 Hoisting, Rigging and Scaffolding

All hoisting, rigging and scaffolding required/associated with the Works described by this specification shall be allowed for by the HVAC contractor.

All scaffolding and elevated platforms to meet with the requirements of the OHS Act, Construction Regulations and Health and Safety procedures.

2.13 Escalation, Forward Cover and Cost of Imported Equipment

The Contractor shall note that this is a fixed price contract.

3 FILTER BOXES

3.1 Filters

Primary filters shall be 50mm thick disposable type panel filters with minimum EU4 efficiency rating and minimum dust holding capacity of 400g/m².

Differential pressure gauges of the magnehelic type shall be provided for each filter bank. Use DWYER or similar.

3.2 Filter Plenum / Mixing Box

Mixing boxes shall be manufactured in 60mm EPS ISO panel with access door for changing filters. A modulating fresh air intake damper, sized according to the required fresh air capacity of each unit shall be provided.

3.3 Air Dampers

Dampers shall be of the opposed blades of extruded aluminium, linked externally with nylon gear wheels. A single drive shaft with Open/Close spring return actuator shall be provided for each fresh air intake. Damper and Actuators shall be Electrovent or similar approved.

4 SPLIT TYPE AIR CONDITIONERS

Supply and install new split type air conditioning unit(s) of the type and capacity as indicated on the drawings.

Only unit(s) supplied by Samsung / Alliance / Midea / Daikin will be considered.

All units shall preferably be of the non-inverter Heat Pump type and utilize R32 / R410a as refrigerant gas.

Each unit shall be provided with an insulated uPVC drain to remove condensate from the building to the nearest drain point. Condensate shall be drained to the nearest drain point at a slope not less than 1:50. All condensate piping shall be provided with a 50mm P-trap.

Special allowance shall be made to prevent condensate dripping from condensers during heating cycles and condensation of refrigeration piping due to high ambient wet-bulb temperatures.

Special allowance shall be made to install refrigerant piping, electrical wiring and condensate piping in PVC trunking where exposed to the outside environment and heavy-duty galvanized wire baskets internally.

Condensers for the split units shall be wall mounted on purpose made galvanised wall brackets or mounted on 600x600 concrete pavers as indicated on the drawings. Exact positions and heights shall be verified on site in conjunction with the Engineer.

The contractor shall allow for drilling of holes through walls and making good after installation.

The contractor shall be responsible for the final electrical connection of each split unit.

Wired remote controls shall be mounted next to each indoor unit with final positions to be agreed on site. Control wiring in shall run in PVC conduit.

Full technical details of the proposed unit(s) shall be submitted to the Engineer for approval before placing any order with the supplier(s).

5 HEAT RECOVERY VRF SYSTEMS

Only unit(s) supplied by Samsung / Daikin / Midea will be considered.

All units shall be of the Heat Recovery type and utilize R410a as refrigerant gas.

Each indoor unit shall be provided with an insulated PVC drain system to remove condensate from the building to the nearest drain point. Where units are installed above ceilings condensate shall be drained to the nearest drain point at a slope not less than 1:50. All condensate piping shall be provided with a 50mm P-trap.

Special allowance shall be made to prevent condensate dripping from condensers during heating cycles and condensation of refrigeration piping due to relatively high internal wet-bulb temperatures.

Special allowance shall be made to install refrigerant piping, electrical wiring and condensate piping in enclosed galvanised wire trays or galvanised trunking where exposed to the outside environment and heavy duty galvanized wire baskets internally.

5.1 Capacities

	Technical Requirements
	VRF1
Total Cooling Capacity (kW)	56 kW (nominal)
Heating Capacity (kW)	63 kW (nominal)

Technical Requirements	
VRF1	
Outside Design Conditions (Summer)	28°C db / 22°C wb
Outside Design Conditions (Winter)	4°C db
Room Design Temperatures	22°C and 55% RH

5.2 Indoor Units

Capacities and type of indoor units shall be as indicated on the tender drawings. Contractors shall allow for all sundries such as hangers and also co-ordinated the positions of units with all other trades on site.

5.3 Multi-port Distribution Boxes

The Contractor shall allow for the necessary multi-port boxes as per the specialist supplier design layout.

5.4 Refrigeration Piping

The refrigeration circuits shall be installed in according to the sizing and the manufacturer's specification.

5.5 Testing and Commissioning

The Contractor shall allow for the complete testing, commissioning and certification of the VRF system by the manufacturer's authorised representative. A full commissioning and test report shall be provided.

6 **OUTSIDE AIR SUPPLY AND EXTRACT VENTILATION SYSTEMS**

6.1 Supply and Extract Ventilation

The contractor shall supply and install supply and extract ventilation systems complete with fans, sound attenuators, sheetmetal and flexible ducting, weather louvers, extract grilles and air valves and electrical connections as indicated on the drawings.

The fan type and minimum air supply capacity of each fan is indicated on the drawings.

Sound attenuators shall be selected to ensure that the maximum sound pressure level of NC 35. Fan speed shall generally not exceed 1450 rpm.

Where fans are not motion sensor controlled/switched, the Contractor shall provide a Stop/Start station for each fan system. The position shall be determined on site (not more than 3m from fan) and the Electrical Contractor shall provide the necessary conduit and draw box / draw wire from the isolator to the Stop/Start station and the fan terminal box. The Contractor shall electrically connect the fan via the stop/start station to the isolator.

6.2 Control of Supply Air Fans

Fan speed shall be automatically adjusted to compensate for filter loading via pressure transducer input directly to EC motor control terminals.

7 DUCTWORK

Air Conditioning and Ventilation ducting shall be manufactured, installed, tested and balanced in accordance with the following specifications, modified as described:

7.1 Manufacturing and Installation

Air Conditioning and Ventilation ducting shall be manufactured, installed, tested and balanced in accordance with the following specifications, modified as described:

- SMACNA: HVAC STANDARD METAL & FLEXIBLE - 3RD EDITION (2005)
- SABS 1238 - 1979: AIR CONDITIONING DUCTWORK
- SABS 1073 - 1980: THE INSTALLATION, TESTING AND BALANCING OF AIR CONDITIONING DUCTWORK
- **(The SANS specifications will apply in cases where the SMACNA specification is silent with regard to ducting, fittings and other low-pressure air handling equipment.)**

7.2 Thermal Insulation

Thermal insulation shall be fitted to all air conditioning supply and return air ductwork. Ventilation ducting shall remain uninsulated. All duct type return air plenums only shall be internally insulated with 25mm sonic-liner.

All sheetmetal ducting shall be externally insulated with three (3) layers of foil-faced bubble wrap fixed/secured onto the duct surface with contact adhesive and nylon strapping at every 1000mm interval.

All flange joints shall be suitably sealed (gasket) and insulated with minimum 150mm wide bubble wrap strips and nylon strapping on either side of the flange and the edges taped with duct tape.

Any delamination of insulation due to air leakage shall be corrected and the insulation redone at the Contractor's expense.

Special care shall be taken not to damage the external foil facing in all areas where ducting will be exposed/visible. Any substandard or damaged insulation shall be removed and redone until an acceptable finish is achieved. The Engineers decision in this matter shall be final.

7.3 Sizing of Ducting

For constant volume systems equal friction method is used. Before manufacturing of the ducting, all drawings with duct sizes shall be sent to the engineer for approval. Wrong size ducting that has been manufactured without the approval of the engineer will be for the contractor's account.

Sizes indicated on the drawings are sufficient for pricing purposes. The Contractor shall verify each duct system with the Engineer (obtain approval) before commencing with manufacture of ducting.

7.4 Sound Attenuators

Sound attenuators of the length and dimensions indicated on the drawings shall be supplied and installed. Final selection of sound attenuators shall be done in conjunction with the fan supplier in order to ensure compliance with acoustic requirements of the building.

7.5 Sealing and Painting of Ducting

All external MEZ flange joints shall be sealed and waterproofed.

No painting of ducting is envisaged.

7.6 Flexible Ducting

All flexible ducting shall have a class 1 fire rating. Certification documentation shall be provided on request by the Engineer.

Sizes shall be as indicated on the drawings. Where no size has been indicated the applicable neck size of air diffusion equipment shall be used. Flexible duct shall be securely fixed onto starter spigots and diffuser sleeves by suitable stainless steel strapping and strapped with duct tape.

Where indicated on the drawings flexible supply air ducting shall be of the acoustic type similar or equal to AdvantageAir's acoustiflex/whisperflex. Air Conditioning supply and return air flexible ducting shall be insulated. Internal and external jackets shall be of multi layer metal foil type bonded to a helical coil. No PVC components shall be used for any jackets. Insulation shall be fire retardant fibreglass.

Ventilation flexible ducts shall be similar to air conditioning flexible ducts, except that they shall be uninsulated.

8 WEATHER LOUVRES, AIR GRILLES AND DIFFUSERS

N/A

9 PAINTING

N/A

10 ELECTRICAL

10.1 General

All electrical work required for this installation as described in the tender document shall form part the scope of work and the cost thereof shall be included in the tender price.

The works shall include the supply, delivery, installation, testing, commissioning and maintenance of the electrical installation required by the new installations for its successful operation in terms of this document. The works shall be complete in all respects and include every item of equipment and all accessories and services required unless specifically excluded.

Should there be any ambiguity in this part or, between this part and the rest of the document tenderers shall apply to the Engineer for a written ruling clarifying the same. If no ambiguity is pointed out, it shall be taken that none exists.

The following will be provided by others:

N/A

The electrical work associated with this contract can be summarised as follows:-

a) Supply of suitable rated PVC/SWA/PVC/PVC cabling and earth to the following equipment from the associated Control Panels or relevant isolators:

- 5x Split units

- b) Connection to isolators, including cabling and terminations of all HVAC equipment.
- c) Supply and installation of new galvanised wire baskets, supports and hangers from the Control Panel and HVAC equipment for all refrigerant piping and electrical cabling.
- d) PVC conduits including draw boxes and draw wires for control cabling. No control cabling shall be strapped to other services and shall be run in conduit throughout.
- e) Connecting and "proving" satisfactory operation of all items of plant and equipment.
- f) Complete earthing system associated with all items of plant and equipment.
- g) Commission and test the entire installation.
- h) Submission of detailed wiring diagrams for approval before alteration/manufacture of any Control Panels.
- i) Provide "as-fitted" details, manufacturer's detail drawings schematic/flow diagrams and maintenance manuals.
- j) The complete control system to enable the installation to function as required in this document.
- k) Supply of an Electrical Compliance Certificate on completion of the installation.

10.2 Fire Interlock

A fire signal shall be supplied to all Control Panels by others. The Contractor shall include in the control layout for a fire relay and wiring thereof to shut down all supply air fans and the activation of any smoke extract system (with auto reset) until the fire alarm has been reset on the fire panel.

11 **CODING, LABELLING AND NOTICES**

11.1 General

The Contractor shall supply and install all coding, labelling and notices as required under this Clause.

The wording shall be in English.

To reduce the possibility of incorrect labels and/or notices, the Contractor shall submit a schedule of labels and notices to the Engineer for approval. Costs to rectify inscriptions, resulting from the failure by the Contractor to obtain approval, will be for his account.

11.2 Coding

General

For painting colour coding refer to the specific table included in this document.

- a) Codes and numbers for wiring shall be CRITCHLEY IZ-type Cable Marker interlocking endless expanding markers, as supplied by CABLE ACCESSORIES (PTY) LTD. CRITCHLEY C-type Cable Markers shall only be used with the approval of the Engineer where wires and piping have already been terminated.
- b) Lettering shall be marked in black on a white background.

Electrical

Provide and install the following coding:

- (a) Numbering of both ends of power and control conductors in switchboards.
- (b) Numbering of both ends of field cables.
- (c) Numbering of both ends of individual field conductors within cables of control circuits only, where such conductors are not uniquely identified by means of insulation colour codes.

11.3

Labelling

General

- a) Labelling shall be CRITCHLEY UNILABEL Cable Marker, as supplied by CABLE ACCESSORIES, or engraved "IVORENE" or "TRAFOLITE" labels.
- b) Black letters on a white background shall be used.
- c) Labels shall be fixed with screws.

Equipment

- a) All mechanical, electrical and instrumentation equipment shall be identified by means of an equipment code.
- b) Minimum height of letters: 10mm
- c) Flow arrows shall be suitably affixed to all pipework wherever required for suitable identification of flow.
- d) All valves shall be uniquely identified with discs and the identification referred to on the "as-built" drawings. Discs shall be brass, aluminium or plastic, not less than 30mm diameter fastened to the valve with stainless steel wire 2mm diameter.

Electrical

Provide and install the following labels:

- a) Code to identify switchboard. Minimum height of letters: 32mm.
- b) Description to identify each outgoing control circuit. Minimum height of letters: = 4mm.
- c) Description to identify function of equipment on the outside face of the switchboard.
- d) Equipment code inside switchboard. Minimum height of letters: 10mm.
- e) Rating of fuses inside switchboard. Minimum height of letters: 10mm.

11.4

Notices

Supply and install all notices required in terms of Statutory Regulations.

In terms of the Occupational Health and Safety Act, at the entrance to each plant room or plant yard, the following notice shall be provided:

- a) Prohibiting unauthorised persons from entering.

- b) Prohibiting unauthorised persons from handling or interfering with electrical apparatus.
- c) Directions as to procedure in case of fire.
- d) Directions as to resuscitation of persons suffering from the effects of electrical shock.

12 CONTROLS SPECIFICATION

12.1 Split Units

A wired remote controllers shall be installed for each system.
Each system shall be switched On/Off via a remote (illuminated) push button located in each respective lecture venue.

The push button control shall activate a rundown timer (adjustable) to automatically switch off the unit after the set time period.

Each unit shall be integrated with the BMS system.

12.2 Temperature Control

Heating and Cooling changeover shall be automatic.

Return air temperature control shall be used.

Temperature setpoint and control parameters shall be programmed as specified below:

Temperature setpoint	:	22,0 °C
Deadband	:	±1,0 °C
Cooling/Heating differential	:	±2,0 °C

12.3 BMS and Control Upgrade

The current BMS shall be upgraded to include for return air CO2 sensing, fresh air damper control, unit On/Off and variable timer control.

BMS shall indicate status of unit, fault and operating parameters as listed above.

13 SERVICE AND GUARANTEE

- General Requirements:

The Contractor shall provide all services, materials and equipment necessary for the successful operation of the entire control system for a period of one year after completion of the successful performance and validation testing resulting in the issuing of the practical completion certificate.

- Description of Work:

The adjustment, repair or replacement of the control system components including all electronic equipment, software updates, transmission equipment and all sensing and control devices.

- Personnel:

The Contractor shall provide qualified personnel to accomplish all work promptly and satisfactorily. The client shall be advised in writing of the name of the designated service representative, and of any changes in personnel.

- Schedule of Work:

Provide four (4) major inspections at 3 monthly intervals.
Clean all equipment, including interior and exterior surfaces.
Perform signal, voltage and system isolation checks.
Check and calibrate each field device.
Check all analogue points and digital points.
Run all diagnostics and correct all previously diagnosed problems.

- Emergency Service during Maintenance and Guarantee period:

The Employer or the Engineer will initiate service calls when the system is not functioning properly. Qualified personnel shall be available to provide service on the complete system. The client shall be given all the relevant telephone numbers where the service representative(s) can be reached at all hours. A service representative shall attend to the call-out on the same day. The control system shall be restored to the proper operating condition within 48 hours.

- System Modifications:

The Employer and Engineer shall be provided with any recommendations for system modifications in writing. No system modifications, including operating parameters and control settings, shall be made without prior approval of the Employer/Engineer. Any modifications made to the system shall be incorporated into the operations and maintenance manuals, and other documentation affected.

14

TRAINING

The Contractor shall provide competent instructors to give full instruction to designated personnel in the adjustment and operation of the system. Instructors shall be thoroughly familiar with all aspects of the subject matter they are to teach.

All training shall be held during normal working hours of 9:00 a.m. to 4:30 p.m. weekdays as follows:

Provide eight (8) hours of training for Employer's personnel. Training shall include:

- Explanation of drawings, operations and maintenance manuals.
- Walk-through of the job to locate control components.
- Operator functions and adjustments
- Explanation of adjustment, calibration and replacement procedures.

Provide four (4) hours of additional training, quarterly for a period of one year from completion of the project.

PART C4: ANNEXURES

UNIVERSITY OF FORT HARE FINANCIAL OFFICE - HVAC UPGRADE

APPENDIX A: DRAWINGS

Drawing Register:

DRAWING NO.	TYPE	DESCRIPTION
P8080M02-001	A1	HVAC LAYOUT